Vexoo Labs – AI Engineer Assignment

Objective

The objective of this project is to design an end-to-end AI pipeline that transforms unstructured documents into structured, queryable knowledge and enables efficient semantic retrieval. The focus is on system design, modularity, and reasoning rather than heavy computation.

Document Ingestion & Knowledge Pyramid

To process large documents, a sliding window approach with overlap was implemented. This ensures that context is preserved across chunks instead of losing important information at boundaries.

A hierarchical Knowledge Pyramid was designed with four layers:

- Raw Text: Original document chunks
- Summary: Condensed representation of each chunk
- Category: Rule-based classification (technical, AI, general)
- Keywords: Distilled important terms from text

This layered structure helps in organizing unstructured data into meaningful and searchable knowledge.

Retrieval Approach

TF-IDF vectorization was used to convert text summaries into numerical vectors.
User queries are also transformed into vectors and compared using similarity scoring to retrieve the most relevant information.

This approach provides a lightweight and efficient alternative to heavy embedding models.

GSM8K Training Pipeline

The GSM8K dataset was used to simulate a machine learning training workflow.

- Selected 3000 training samples and 1000 test samples
- Implemented data loading and preprocessing
- Simulated training loop with logging
- Evaluated using accuracy metric

Due to resource constraints, training was simulated while maintaining correct pipeline structure.

Design Decisions

- Used TF-IDF instead of embeddings for efficiency

- Focused on modular and scalable architecture
- Prioritized clarity and reasoning over complex implementation

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Bonus: Reasoning Router

A routing mechanism was designed to classify queries based on intent (e.g., math, general, technical) and direct them to appropriate modules. This enables a flexible and extensible system.

Conclusion

This project demonstrates the ability to design scalable AI systems, structure knowledge effectively, and implement end-to-end pipelines with clear engineering decisions and trade-offs.